

Martha Nwanua

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ABOUT ME

ASCP-eligible, HCPC Registered Biomedical Scientist with a strong drive to ensure the growth and development of any organization using the application of knowledge and experiences gathered. I possess over three years of impeccable clinical laboratory skills in all fields of Medical Laboratory Science (including Clinical Chemistry, Microbiology, Hematology, and Histopathology) with additional skills in molecular biology, Inventory, public health advocacy, quality management system, effective leadership, and research management. I have experience working in BSL 2 and 3 Laboratories processing AFB and COVID-19 samples in compliance with the ISO 15189 standards. Furthermore, I have excellent communication, good analytical, and interpersonal skills with the ability to perform with a high degree of professionalism at all times. I have a keen interest in Molecular biology, Cancer awareness, HIV/AIDS advocacy, Public health, and Good Clinical Laboratory Practice.

Registration number: **BS078148**

SKILLS

Laboratory Skills

- CLS Skills and CLIA
- Hematology and Blood Bank Skill
- Patient Care Skill
- Quality Control and Quality Management System (Q-pulse, Winpath, APEX, and Telepath)
- Clinical Chemistry Skill
- Molecular Biology (RT-PCR, qPCR, GeneXpert, HPLC & Electrophoresis)
- LMIS
- Phlebotomy.
- Instrumentation and Preventive Maintenance
- Microbiology

EDUCATION AND TRAINING

Bachelor of Medical Laboratory Science (BMLS)

Nnamdi Azikiwe University, Awka, Anambra State. [09/2013 – 07/2018]

Country: Nigeria

Field(s) of study: Biomedical Sciences

Final grade: 3.46/5.00

Thesis: Effect of Coffee Consumption on Plasma Glucose and Lipid Profile Level in Male Student at Nnamdi Azikiwe University. Grade :B

- Chemical Pathology, Therapeutic drug monitoring and Toxicology, Genomics and Bioinformatics, Biomedical Engineering, Cell Biology, Embryology, and Genetics, Humoral Immune Response, Virology, Metabolism, Enzymology, Cardiovascular Disease, Pharmacology, and Microbiology.

WORK EXPERIENCE

Specialist Biomedical Scientist

APIN Medical Laboratory and Diagnostic [01/2022 – Current]

City: FCT

Country: Nigeria

- Perform all blood sciences tasks to support the specialized diagnostic service for patients and work on a rotational basis through all sections of the department including chemistry, hematology, blood bank, and transfusion,
- Execute analysis of the concentration of substances such as glucose, lipids, enzymes, electrolytes, hormones, proteins, and other metabolic products present in human blood and urine using manual and automated methods
- Execute hematological tests including red and white blood counts, sickle cell tests, hemoglobin estimation, fragility tests, coagulation time, and clot retraction time.
- Perform clinical genetic procedures in molecular and cytogenetics on an extended bench rotation following established protocols such as in-situ hybridization procedures, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) extraction, and quantification, polymerized chain reaction and reverse transcriptase (RT)-PCR analysis, culture, harvest, preparation and chromosome analysis of bone marrow, blood, and solid tumors, and preparation of karyotypes.
- Perform biochemical testing on proteins and antibodies including, but not limited to, CE, HPLC, SEC, and UV, ELISA techniques, HPLC/FPLC and Affinity Purifications, SDS-PAGE, CGE, Isoelectric Focusing, and Blotting techniques.
- Report, analyze, review, and interpret External Quality Assessment data, particularly those for POCT methods. Troubleshoot poor EQA returns including both technical and operator-related problems. Report EQA performance at the weekly team meeting.
- Process and validate patient specimens analyzed on the Cobas 8000 line, Cobas, e411 immunochemistry analyzer, Cobas c311, and C11 automated clinical chemistry analyzer, Beckman Coulter hematology analyzer, Sysmex BC 5300 5-part hematology analyzer, Roche H800, Cobas X480, Partec, Mindray BC 5300, and Mindray 6800 "6 part hematology analyzer, Menarini, Serbia, and Hydrasis analyzers.
- Perform Blood grouping, antibody screening, antibody identification, crossmatching, direct antiglobulin testing, and phenotyping on patient samples.
- Responsible for the performance of compatibility testing and the issue of Blood and Blood Products to patients in routine and emergencies.
- Manage own workload making short-term adjustments to reprioritize workload for urgent specimens and make critical diagnostic decisions unsupervised, and with no on-site backup or reference point when working alone, on-call duty, and on weekends.
- Use competently the laboratory computer systems for patient test results, database entry and retrieval, and equipment some of which are highly automated and expensive or of a specialist nature.
- Taking direct responsibility for all the results produced and work performed including quality control and assurance, assisting with the training of other laboratory staff.

Specialist Biomedical Scientist

Body Affairs Diagnostic Limited [01/2022 – 08/2022]

Country: Nigeria

- Assessed Specimens in areas of hematology, urinalysis, coagulation, and clinical chemistry within time constraints in a high volume diagnostic laboratory.
- Processed biological samples including blood, serum, cerebral spinal fluid, urine, feces, and tissues, taking responsibility for the quality and authorization of the results, which was used in the diagnosis, treatment, and management of patient care.
- Supervised the accuracy and precision of laboratory investigations within the blood sciences department using appropriate quality control procedures for blood transfusion, and hematology.
- Conducted hands-on wet chemistry techniques to support assay development activities and performed toxicology and chemical testing on laboratory specimens, using manual and automated techniques, meeting turnaround times,
- Executed and interpreted the Coagulation test (PT, APTT, DVT, AP50, PT50, Fibrinogen, and D-dimer), interpretation (activator deficiency or clotting factor deficiency), and interferences (Heparin and Warfarin administration).
- Performed moderate to high complexity PCR-based assays on multiple platforms to monitor and detect various Infectious (HIV, HCV, and HBsAg viral load quantification) and executed protocols for DNA/RNA extraction on 3000+ patients samples; SARS CoV-2 PCR antibody test processing; Extraction, Amplification, Pipetting, and aliquoting techniques to minimize cross-contamination

- Operated and demonstrated specialized knowledge of Thermo Quant Studio 5 PCR, DA 7600 PCR amplification machine, Smartez automated extraction PCR machine, Abbot Architect, ELISA Reader ADX-110, Maglumi 600, Sysmex XS-500i CBC analyzer, Sysmex CA-104 Coagulation analyzer, Roche H202 POCT, BD FACS Presto CD4 analyzer, Bio-Rad D10 HPLC and BD BACTEC FX40
- Carried out first-line equipment installation/implementation/training and/or validation/verification and method troubleshooting; advised POCT Co-ordinator of situations requiring further action, and worked with service engineers on fault resolution which may include dismantling instruments to carry out initial repairs, and evaluate the efficacy of repairs.
- Undertook technical validation, interpretation, and authorization of a wide range of complex laboratory results and made decisions that require referral to a consultant or Clinical Scientist
- Liaised with the team manager and laboratory manager, I was responsible for the day-to-day supervision, education, and training of trainee BMSs and MLA

Specialist Biomedical Scientist

State House Clinic [01/2021 – 12/2021]

Country: Nigeria

- Performed laboratory testing as a generalist in Hematology, Serology, and Chemistry, blood bank as well as molecular biology and -experienced working out-of-hours / on-call in a Blood Sciences Laboratory
- Undertook manual, semi-automated, and fully automated laboratory investigations on a rotational basis within the Blood Science Department.
- Performed point-of-care testing of sodium, potassium, chloride, urea (or blood urea nitrogen), creatinine, glucose, hematocrit, and hemoglobin, as well as ionized calcium, lactate, prothrombin time, cardiac troponin, and pH, Po₂, and Pco₂.
- Worked with ward staff to support their POCT needs e.g. training, troubleshooting, need for devices, communication of the benefits and limitations of POCT equipment
- Undertook technical validation of multiple complex sets of results from laboratory investigations ensuring accuracy and precision as specified by laboratory protocols. I also assessed the validity of the results as to whether conflicting information is due to the pathology of the patient, contamination, or compromised integrity of the sample before the release of the results for clinical validation.
- Performed complex tests and assays according to established procedures and required standards. These include microscopic analyses and the plating-out and interpretation of culture plates from infected clinical material such as urine, sputum, and feces and the assaying of blood samples for indicators of infection, including high-risk samples
- Worked in Category 2 (high-risk) conditions as part of duties. Handled potentially high-risk specimens (including blood-borne viral infections e.g. Hepatitis, HIV, nvCJD, TB).
- Responsible for the performance of compatibility testing and the issue of Blood and Blood Products to patients in routine and emergencies.
- Assisted in the maintenance of the Hospital Blood Banks and the stock control of Blood & Blood Products. Provided blood components and products to clinical providers
- Maintained and troubleshooted the Clinical Biochemistry LIMS database and instrument analyzer interfaces. Ensured that appropriate preventative maintenance schedules have been carried out on complex laboratory equipment before use.
- Performed all specimen handling, specimen manipulation, reagent preparation, test system maintenance, chemical/biologics handling, and testing procedures in a manner that ensured the safety of self and others.
- Prepared the statistical charts for review by the Point of Care testing coordinator and the Director and notified the Point of care testing coordinator of remedial actions and follow-ups for all point-of-care tests and instruments.
- Monitored the use of reagents, consumables, and equipment within the blood sciences and ensured stock levels are maintained to ensure continuous service.
- Participated in quality improvement activities such as reviewing cases with abnormal results to assess causes and contributing to corrective actions where applicable.

Trainee Biomedical Scientist

LAB360 International Limited [08/2020 – 12/2020]

Country: Nigeria

- Responsible for the day-to-day processing and validation of results from patient samples: blood, urine, feces, tissue biopsies, CSF, and other body fluids in specialist and routine laboratory sections in Biochemistry using a range of scientific measurements and technical procedures.
- Performed specialized diagnostic and monitoring procedures routinely performed in the laboratory using my discretion whilst following broad policies and procedures.
- Performed audits of working practices and report audit findings and raise non-conformances and observations using Q pulse quality management resource.
- Liaised professionally with other staff in Pathology and handled inquiries from external users of the service appropriately. Provided and received complex information and provided advice, an explanation of results, or instructions,
- Performed instrument maintenance, reagent preparation, and calibration by performing various routine instrument checks, cleaning procedures, and replacement of instrument parts and investigating basic instrument malfunctions on analyzers according to a written schedule or verbal instructions.
- Performed complex tests and assays according to established procedures and required standards. These include microscopic analyses and the plating-out and interpretation of culture plates from infected clinical material such as urine, sputum, and feces and the assaying of blood samples for indicators of infection, including high-risk samples.
- Organised and prioritized my workload and that of others, taking responsibility for delivering results within agreed timescales and standards within an environment of changing priorities and pressures
- Conducted virological and non-virological analysis of serum samples using both automated systems (DS261 and Cobas e411) and manual method,
- Operated, maintained, calibrated, quality controlled and trouble-shot automated equipments such as Roche H800, Cobas X480, Partec Cyflow CD4, Bact alert, Zeiss Fluorescence light microscope, incubators, and hot air oven.
- Identified pathogenic organism, or markers of the cause of infections (serological tests) and carried out antimicrobial sensitivity tests.
- Performed various routine to specialist test in hematology such as:
 1. Full Blood Count using 5 parts and 6 parts hematology analyzer such Mindray, Sysmex, abacus, and Beckman Coulter.
 2. Coagulation test using Roche Ca 102 and manual method using water bath.
 3. Blood Film Preparation (thick and thin film) and staining of prepared film.
 4. Reporting and interpreting blood films and providing restricted clinical comments and advised as per departmental policy
 5. Screening and identifying compatible blood for patients with complex antibodies.
 6. Special staining techniques for Leukemia diagnosis
- Performed some routine to specialist clinical biochemistry duties such as renal function test, liver function test, lipid profile, immunoassay using various equipments such as spectrophotometer, chromatography systems (HPLC, GCMS, and GC) and ICP/AA.

Trainee Biomedical Scientist

National Hospital Abuja [06/2019 – 07/2020]

Country: Nigeria

- Supervised the accuracy and precision of laboratory investigations within the blood sciences department using appropriate quality control procedures for Blood Transfusion, and Hematology
- Performed qualitative and quantitative tests and examinations using various analyzer and/or manual methods on specimen sources such as blood, urine, and other body fluids
- Liaised with POCT coordinator on procurement of new POCT devices (e.g. undertake evaluation and help write evaluation reports). Helped develop, implement, and maintain standards of practice, policy, and procedure for POCT (e.g. writing and review of SOPs via Q-pulse as directed by the POCT coordinator).
- Ensured compliance with the Trust POCT Policy and UKAS accreditation ISO standards for POCT are met.
- Prepared, inspected, and issued blood and blood products to assure the safety of transfusion to the patient. Strictly adhered to proper unit/patient identification, and determination of safe, appropriate products.
- Assessed the suitability and appropriateness of samples submitted to the laboratory and took remedial action when required.

- Ensured the integrity of the patient database within the laboratory computer system by accurate registration of patient demographics and request details. Input laboratory results generated by self and others.
- Processed samples for Hemoglobin electrophoresis and glycated hemoglobin. Achieved success in providing a solution to an extremely complicated blood test employing insight and relevant electrophoresis knowledge.
- Demonstrated operational knowledge of instruments such as Roche C411, Roche C311, Architect Abbott 4000 and 100ir, Bact-Alert 3D, Bactec 9050, Vitek 2 compact Biomerieux, Sysmex XE 5000, etc.
- Loaded the Bactec instrument for the rapid detection of microorganisms in blood and sub-cultured the blood sample when flagged as positive.
- Performed D-Dimer, Troponin I, clotting profile (PT INR, APTT), Indirect Coombs Test, and Direct Coombs Test on patients' samples..
- Inoculated and incubated culture plates, analyzed cultures, decided on possible significant pathogens, identify micro-organisms, performed antibiotic susceptibility tests, validated and reported results and interpretive comments using judgment in determining and initiating further tests that may be required.
- Performed phlebotomy procedures as assigned to more than 50% of patients in a 14-bed clinic utilizing vacuum tube, syringe, and venipuncture butterfly method to draw blood from veins as well as capillaries through dermal punctures such as heel or finger stick method
- Responsible for the transmission of laboratory test results (paper, electronic, verbally), including the provision of telephoned urgent and emergency results and associated specialist advice relating to the appropriateness of samples and investigation requests.

PUBLICATIONS

Effect of Coffee Consumption on Plasma Glucose and Lipid Profile Level in Male Student at Nnamdi Azikiwe University

[2019]

Ihim Augustine Chinedu, Nwanua Martha Ifeyinwa, Ogbodo Emmanuel Chukwuemeka and Meludu Samuel C.

CONFERENCES AND SEMINARS

Antibody Identification with Helen Thom of NHS Blood and Transplant

[Biomedical Scientist Empowerment, Education and Discussion Group, 2022]

Hemoglobinopathies: Challenges and Considerations for the Hospital Transfusion Laboratory

[Daniel Monteiro, Principal Biomedical Scientist n Special Hematology at Viapath Analytics, 2022]

Transfusion Laboratory Perspective of Autoimmune Haemolytic Anaemia

[Biomedical Scientist Empowerment, Education and Discussion Group/ NHS Blood and Transplant, 2022]

Innovative ESR and Platelet Detection in Clinical Hematology Webinar

[Mindray, 2022]

What does it take to be a good research leader?

[The Global Health Network, 2022]

National Laboratory Quality Framework (NLQF) Workshop

[African Society For Laboratory Medicine, 2022]

Molecular and Serological Testing for COVID-19

[Webinar CME, 2020]

Skill Acceleration BootCamp Soft Skills Training

[Skill Factory Africa, 2020]

HONOURS AND AWARDS

Healthcare Systems Short Course

University of Nottingham [2022]

Basic Molecular Biology Module 4: PCR and real-time PCR

Center for Disease Control and Prevention (CDC) [2022]

Basic Molecular Biology Module 1: Basic Science

Center for Disease Control and Prevention (CDC) [2021]

Introduction to Infection Prevention & Control (IPC)

World Health Organization [2021]

Let's Break the Chain of COVID-19 Infection

Mohammed Bin Rashid University of Medicine and Health Sciences [2021]

Save a Life Certification: CPR, AED & First Aid (infant, child and adult)

National Health Care Provider Solution, Las Vegas [2020]

Certificate of Excellence "Overall Best in Seminar Presentation"

Association of Medical Laboratory Council of Nigeria: National Hospital Abuja Chapter [2020]

Covid-19 Operational planning guidelines and Covid-19 partners platform to support country preparedness and response

World Health Organisation [2020]

Prevention of Hypertension

European Accreditation Council for continuing Medical Education [2019]

Statement of Accomplishment with Excellence in "Partnering with the Public and Patient in Medical Research

Stanford University School Of Medicine [2019]

VOLUNTEERING

Eternity Network International

[POCT Specialist Biomedical Scientist, 02/2020 – Current]

- Collaborated in a team with other healthcare practitioners to provide free medical services and emergency treatments on weekends.
- Perform COVID-19 Antigen test and other Rapid Diagnostic tests.
- Collect and prepare nasal-pharyngeal samples for PCR testing of COVID-19 by the standard method and processes. Transcribe nasal and oral samples for laboratory testing.
- Responsible for maintenance, troubleshooting, and repair of Point of Care Testing (POCT) ward-based analytical systems
- Made technical assessments of method and instrument performance by monitoring quality control used on specialist laboratory methods and analytical systems
- Monitored and controlled ward-based POCT analyzers remotely by using the RapidLink remote control computer system

Health Initiative for Safety and Stability in Africa / ABUJA Medical Fest

[Biomedical Scientist (POCT), 11/2021]

- Participated in the outreach organized by IFASS in conjunction with National Agency for the Control of AIDS to provide free HIV screening and counselling in communities.

Staff Clinic Secretariat, Abuja

[Biomedical Scientist Trainee, 08/2018 – 05/2019]

- Closely followed trends and techniques in medical laboratory testing. Stringently followed lab protocols, sanitation standards and testing procedures.
- Developed essential skills in prioritization and communication to medical staff working in the laboratory servicing the Emergency Department and wards.
- Analyzed samples for exposure to notifiable diseases including Dengue, Tuberculosis, and provided timely results to treating doctors.
- Undertook routine manual analysis for porphyrins, osmolality, renal calculus.
- Routine serological testing, including Antenatal, Hepatitis and HIV screening.
- Haematology: Processed samples through a haematology analyzer, make films and screen films alongside a haematology scientist, Watched performed routine coagulation testing.
- Immunohaematology or transfusion medicine: Undertook group and screening of routine antenatal and urgent blood samples, perform crossmatch using both manual and automated techniques (including Biorad or Grifols), issuing of blood products including packed red cells, platelets and fresh frozen plasma.
- Microbiology: Performed routine screening of urines, faeces and sputums, including plating and microscopy. Watched colonial morphological identification and biochemical identification of pathogens.
- Clinical Biochemistry: Undertook routine biochemistry analysis using an automated biochemistry analyzer Beckman AU680, Roche, Hitachi.
- Histology: watched and performed cut up and embedding, sectioning and staining of tissues. Watched and took part in an autopsy.
- Specimen Reception: Undertook the routine receipt of samples and performed all identification checks, coding and computer entry in a busy laboratory.

PROFESSIONAL MEMBERSHIP

Health and Care Professional Council (HCPC)

[25/08/2022 – Current]

Association of Medical Laboratory Scientists of Nigeria

[25/08/2019 – Current]

American Society for Microbiology

[2021]

ACHIEVEMENTS

Achievement and Accomplishments

- Consistently received commendations for adaptability to changing situations and priorities Skills: Lab Management, Writing (Methods, Test Plans, SOPs).
- Nominee for the Academic Excellence Scholarship for Master in Molecular Oncology (partially-funded).
- Certificate of excellence for the overall best seminar presenter amongst 23 presenters at the National Hospital, Abuja (2020)
- Received a commendation for being the most outstanding staff at Body Affairs Diagnostic, Abuja
- Spearheaded 2023 the External Quality Assurance Scheme for the Blood Science Department with the Medical Laboratory Council of Nigeria for APIN Medical Laboratory and Diagnostics.

DIGITAL SKILLS

Microsoft Office

SKILLS OVERVIEW

Technical Skills

Proficient in the use of:

- Microsoft Office Suite (Excel, Word, Power point)
- Google Suite (Gmail, Docs, Drive)
- LIMS (Palash, Health in the Box, etc)

Operational Skills

- **Chemistry Analyzers:** Roche (C411 and C311), Architect Abbott (4000, 1000ir, Ci4100 and i1000SR), Selectra PRO S, Analyzer ISE 4000, DS 261 semi-automated analyzer.
- **Microbiology Analyzers:** Bact-Alert 3D, Bactec 9050, BD BACTEC FX40, Vitek 2 compact Biomerieux, Centrifuge, Microscope, Memmert oven uf55, Fisher scientific incubator.
- **Histopathology:** Leica TP1020 Tissue Processor, USICO Tissue Embedding Station, Manual Rotary Microtome.
- **Hematology Analyzers:** Sysmex CBC analyzer (XE 5000 and XS-500i), Sysmex CA-104 Coagulation analyzer, Roche H202, BD FACS Presto, Bibby Scientific SSM1 Mini Orbital Shaker, Mythic 22, Abacus 380.
- **Immunoassay Analyzers:** Maglumi 600, Getein 1100, ELISA Reader ADX-110.
- **PCR:** Thermo Quant Studio 5 PCR, DA 7600 PCR amplification machine, Smartez automated extraction PCR machine.
- **Electrophoresis:** Electrophoretic tank, Bio-Rad D10 HPLC.

Soft Skills

- Communication and Interpersonal skill
 - Organizational and planning skill.
 - Problem-solving skill.
 - Good hand-eye coordination.
 - Ability to use technical equipment with accuracy.
 - Excellent attention to detail.
 - Ability to work methodically and meticulously when following scientific techniques and procedures.
 - Ability to work independently and as a team.
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